Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 5

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 3

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

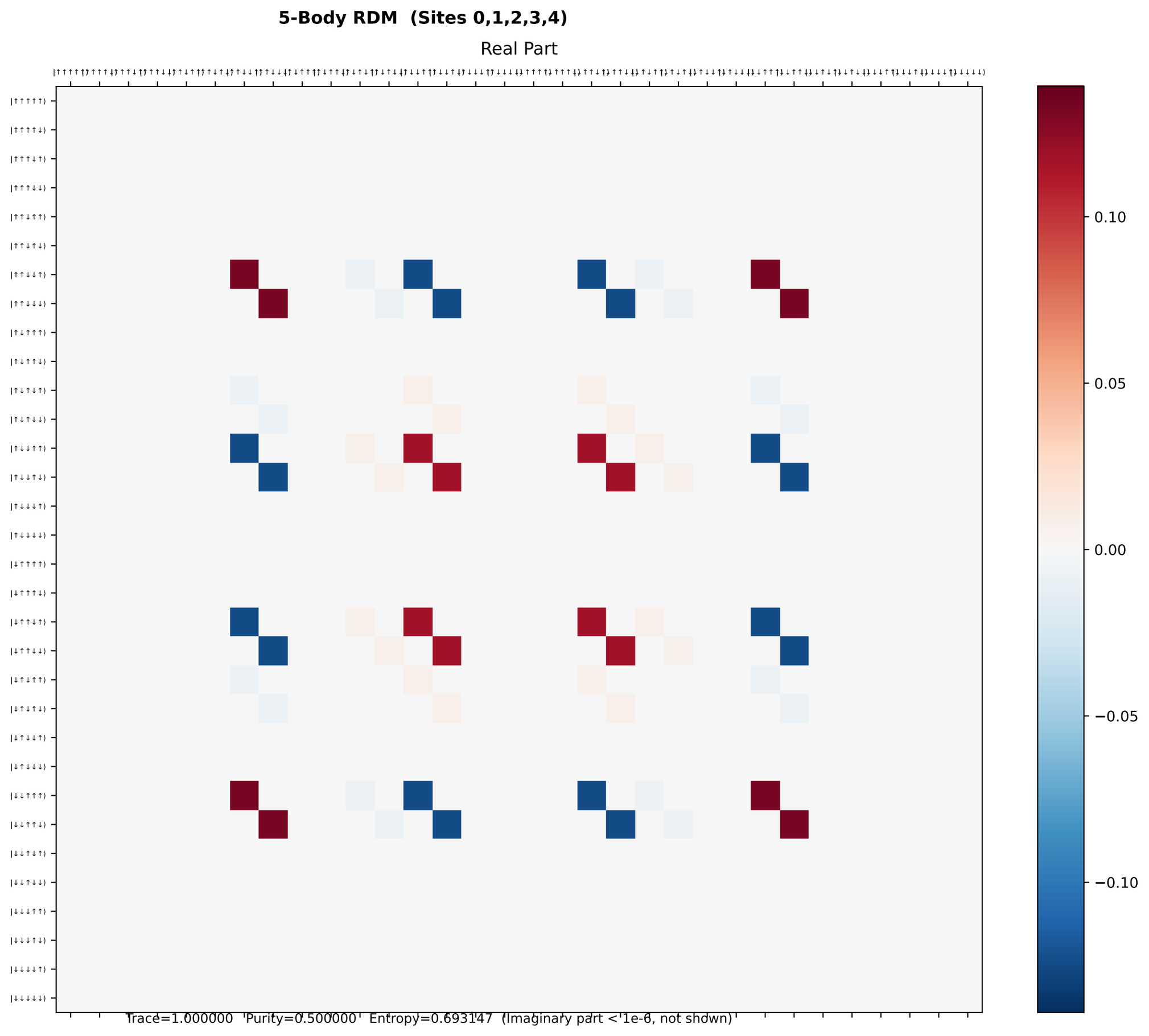
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Body order : 5
Clusters   : 6

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Summary — 5-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2,3,4	0.500000	—	0.000000	0.000000	—	0.000000	MIXED
0,1,2,3,5	0.500000	—	0.000000	0.000000	—	0.000000	MIXED
0,1,2,4,5	0.500000	—	0.000000	0.000000	—	0.000000	MIXED
0,1,3,4,5	0.500000	—	0.000000	0.000000	—	0.000000	MIXED
0,2,3,4,5	0.500000	—	0.000000	0.000000	—	0.000000	MIXED
1,2,3,4,5	0.500000	—	0.000000	0.000000	—	0.000000	MIXED


5-Body RDM (Sites 0,1,2,3,4)
Real Part
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 0,1,2,3,4)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.5000000000
  Von Neumann entropy  : 0.6931471806
  Eigenvalue min       : -1.110978e-16
  Eigenvalue max       : 5.000000e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.50000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

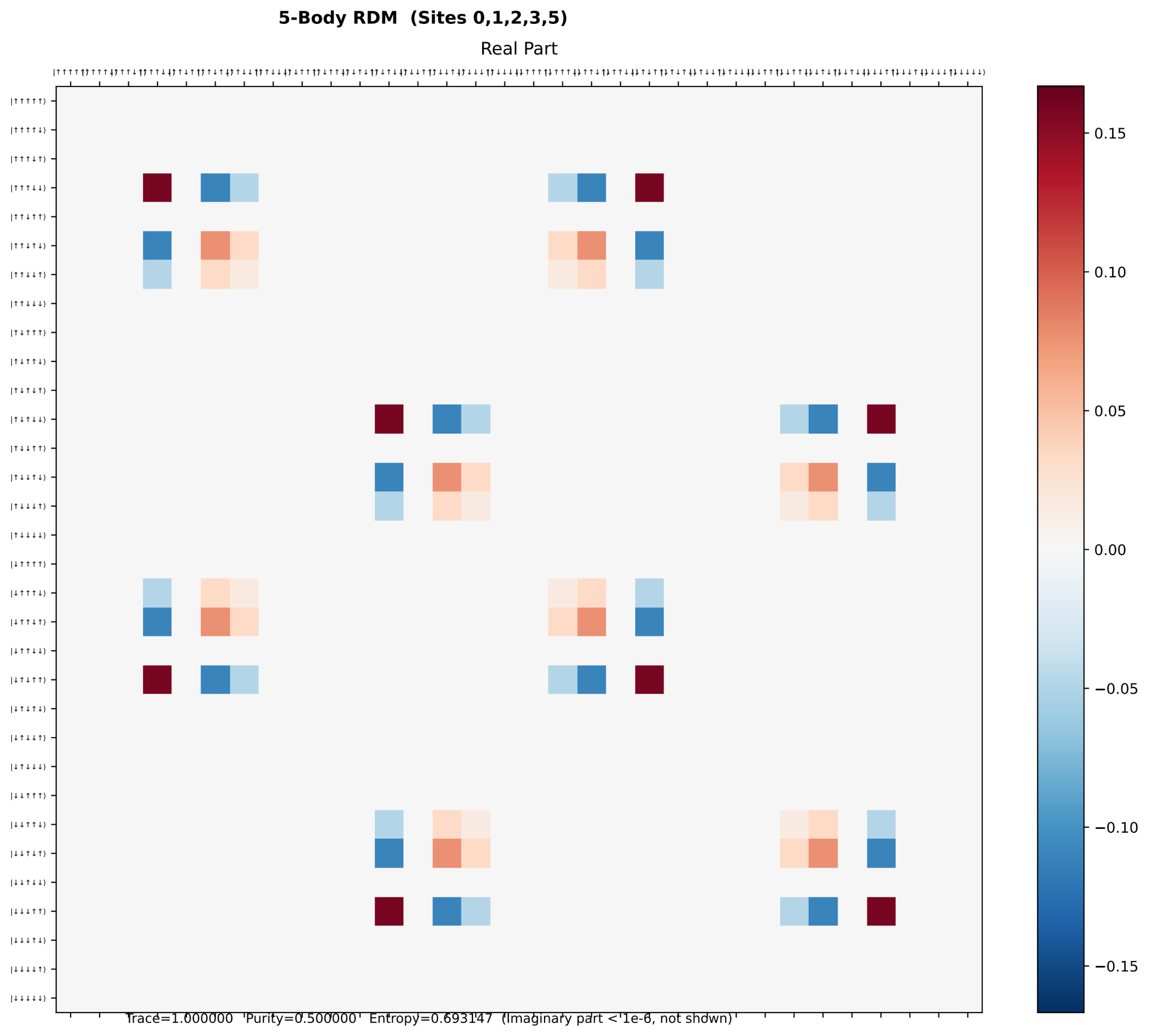
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000004

  Status: no_GME

  Solver source:
    –
```

```
1. p=0.199794  C_GME=0.000000
2. p=0.442785  C_GME=0.000000
3. p=0.357421  C_GME=0.000000
```


5-Body RDM (Sites 0,1,2,3,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <'1e-6, not shdwn)

Analysis Summary - 5-Body RDM (Sites 0,1,2,3,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: -2.267998e-17
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

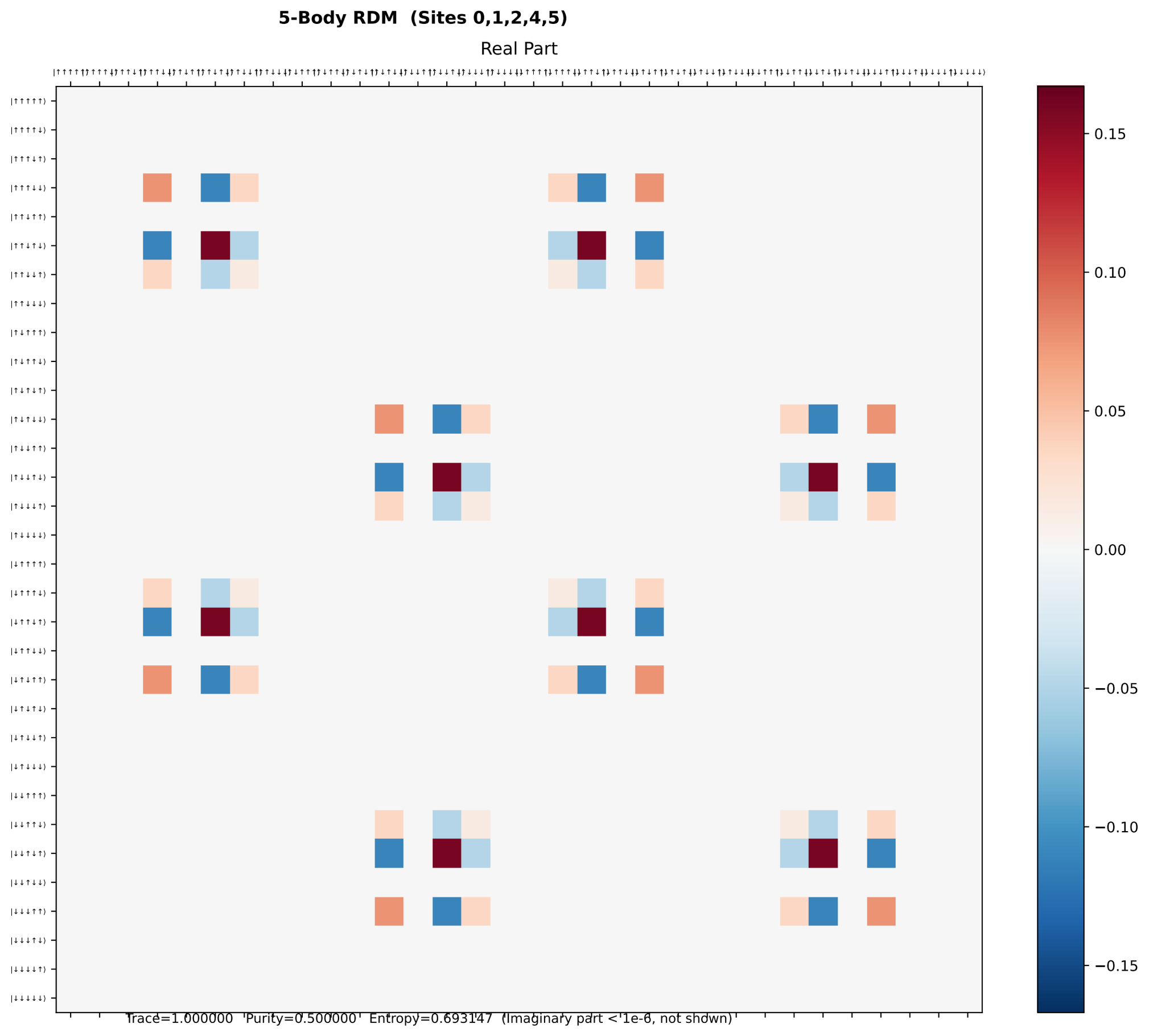
Convex Roof Analysis
==========

Lower Bounds:
  C_LB = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.00000003

Status: no_GME

Solver source:
  –

1. p=0.199792 C_GME=0.000000
2. p=0.442784 C_GME=0.000000
3. p=0.357423 C_GME=0.000000


5-Body RDM (Sites 0,1,2,4,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <'1e-6, not shdwn)

Analysis Summary - 5-Body RDM (Sites 0,1,2,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.5000000000
Von Neumann entropy	0.6931471806
Eigenvalue min	-5.205788e-17
Eigenvalue max	5.000000e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

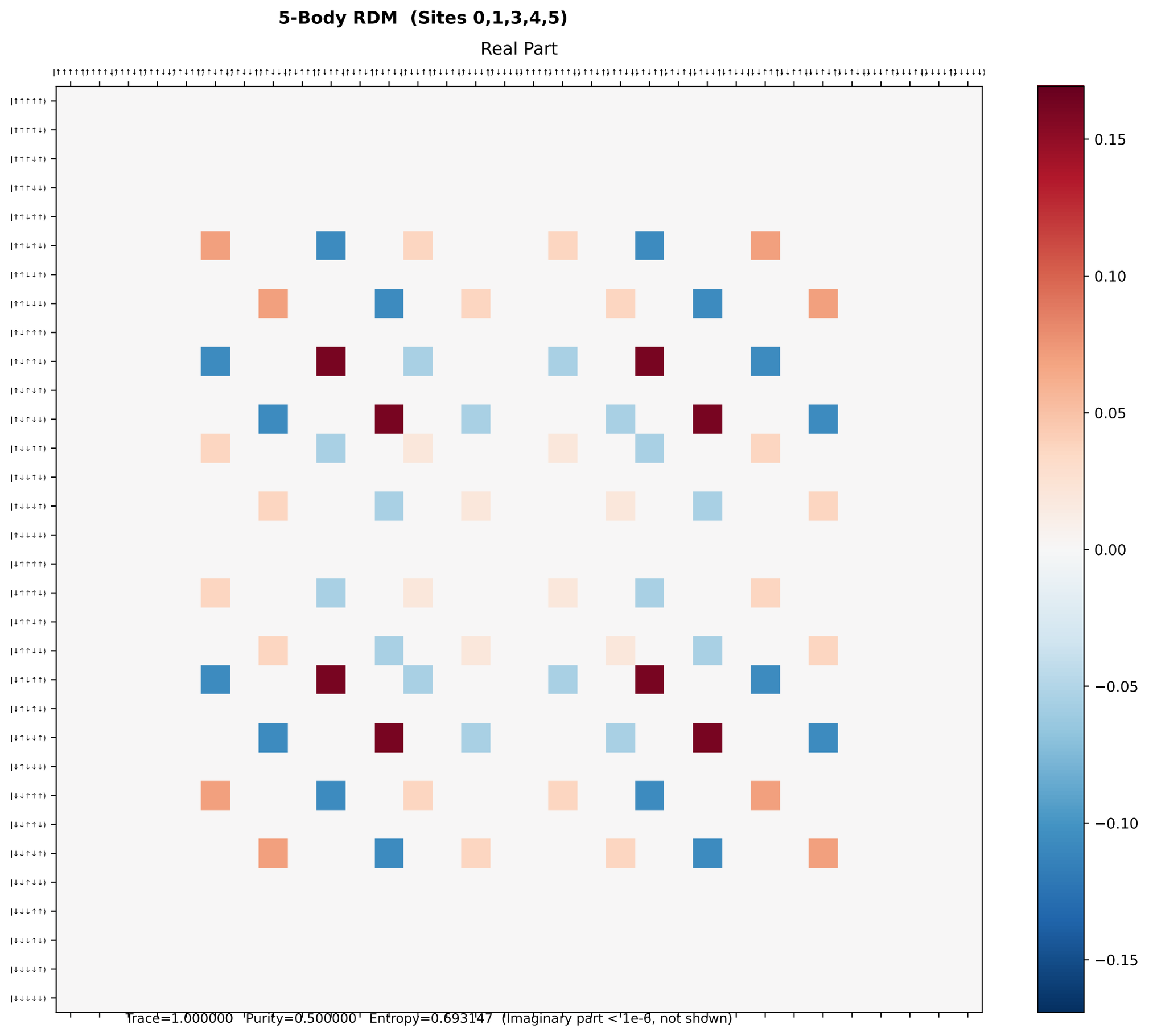
Convex Roof Analysis
==========

Lower Bounds:
  C_LB = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.00000002

Status: no_GME

Solver source:
  –

1. p=0.199792 C_GME=0.000000
2. p=0.442784 C_GME=0.000000
3. p=0.357423 C_GME=0.000000


5-Body RDM (Sites 0,1,3,4,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 0,1,3,4,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.5000000000
  Von Neumann entropy  : 0.6931471806
  Eigenvalue min       : -5.692633e-17
  Eigenvalue max       : 5.000000e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.50000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

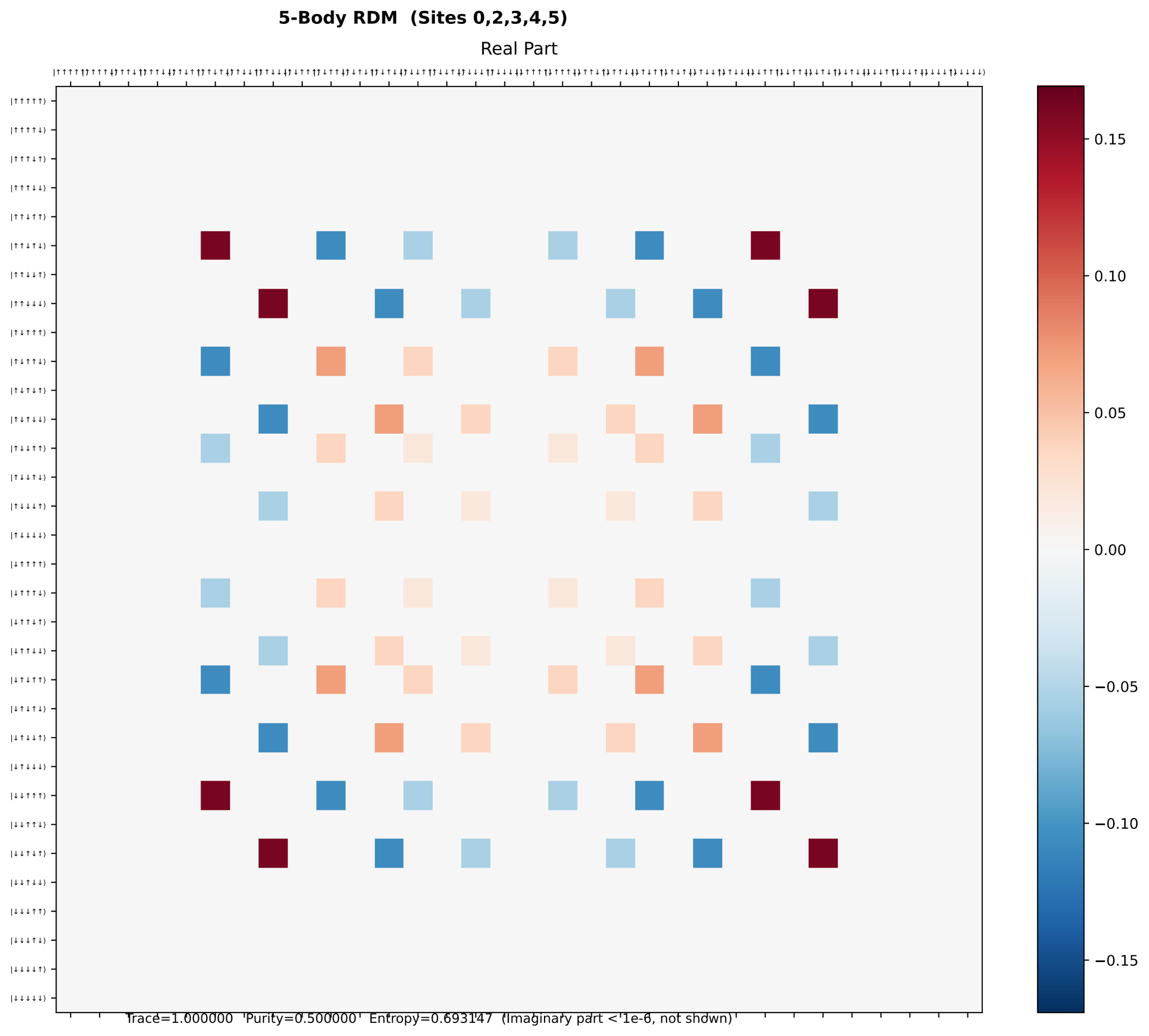
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000003

  Status: no_GME

  Solver source:
    –
```

```
1. p=0.199792  C_GME=0.000000
2. p=0.442784  C_GME=0.000000
3. p=0.357424  C_GME=0.000000
```


5-Body RDM (Sites 0,2,3,4,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 0,2,3,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.5000000000
Von Neumann entropy	: 0.6931471806
Eigenvalue min	: -3.731689e-17
Eigenvalue max	: 5.000000e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

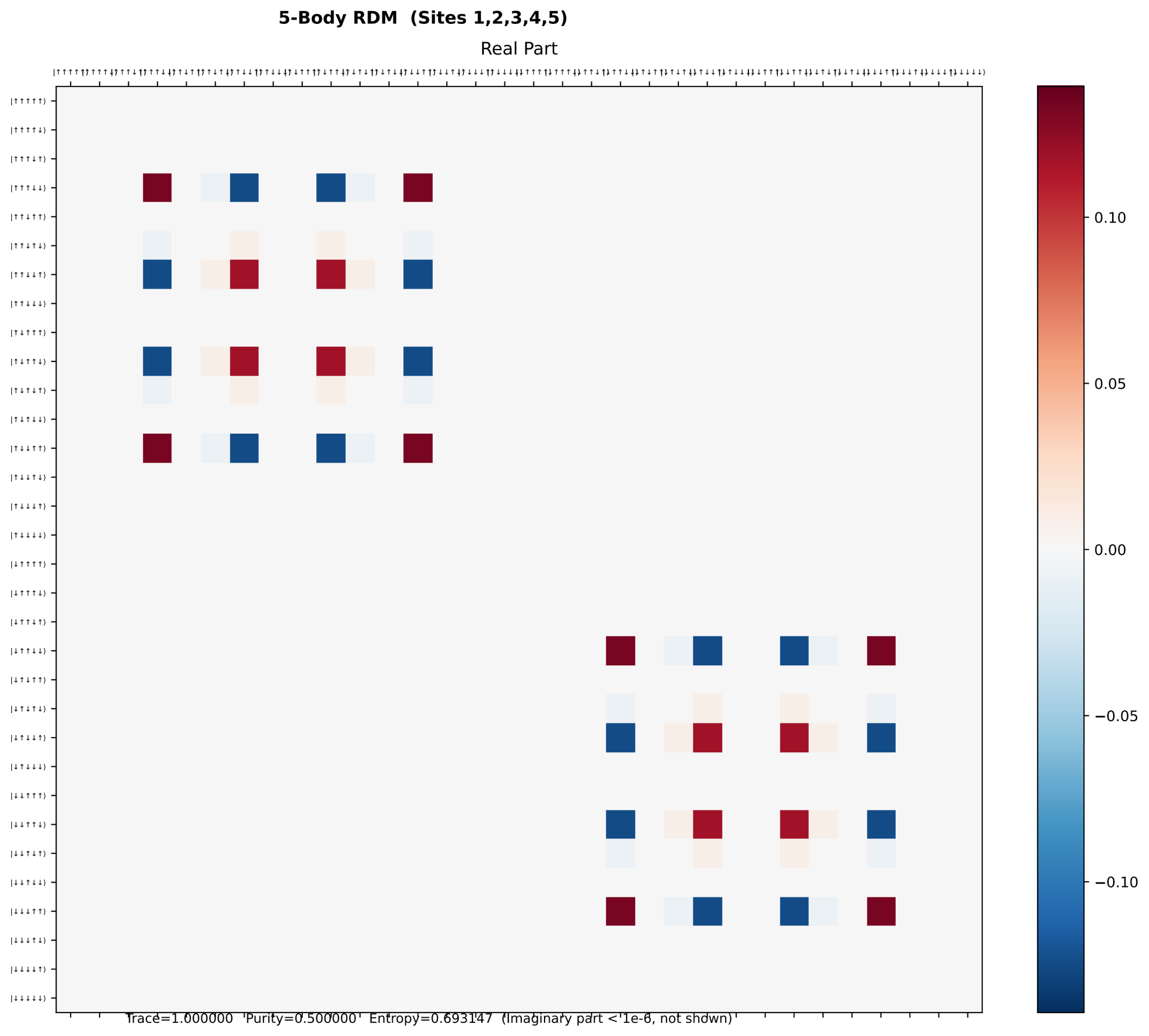
Convex Roof Analysis
==========

Lower Bounds:
  C_LB = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.00000000

Status: no_GME

Solver source:
  –

1. p=0.199792 C_GME=0.000000
2. p=0.442785 C_GME=0.000000
3. p=0.357423 C_GME=0.000000


5-Body RDM (Sites 1,2,3,4,5)
Real Part
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 1,2,3,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.5000000000
Von Neumann entropy	0.6931471806
Eigenvalue min	-4.990995e-17
Eigenvalue max	5.000000e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis
==========

Lower Bounds:
  C_LB = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.00000004

Status: no_GME

Solver source:
  –

1. p=0.199792 C_GME=0.000000
2. p=0.442784 C_GME=0.000000
3. p=0.357423 C_GME=0.000000